BizPilot - Project Bible

> *AI Autopilot for Small & Medium Businesses Last updated: 2026-03-12*

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1. Project Overview

Vision

BizPilot is an AI agent platform that helps small and medium businesses automate customer care and page management across multiple channels (Facebook Page, Instagram, Website Chat, WhatsApp, Zalo...) at a fraction of the cost of hiring staff. Built for any market, with initial focus on US SMEs.

Value Proposition

- **For SMEs**: One AI agent working 24/7, handling 60-70% of page admin work, cheaper than hiring a person
- **For us**: Build once, deploy to any market, high margin, scales globally

Target Markets

Priority	Market	Key Channels	Language	Notes
Primary	United States	Facebook, Instagram, Website Chat, WhatsApp	English	Largest SME market, highest willingness to pay
Secondary	Vietnam	Facebook, Zalo, Website Chat, WhatsApp	Vietnamese	Existing network, initial pilot customers
Future	Southeast Asia, LATAM, Europe	Market-dependent	Multi-language	Expand as platform matures

Target Customers

- SMEs with active social media pages (Facebook, Instagram) selling products or services online
- 50-10,000 products/services
- Industries: e-commerce, beauty/cosmetics, fashion, F&B, home services, fitness, real estate, professional services
- Currently using: manual inbox management, basic chatbots (ManyChat/Chatfuel), or no automation at all
- Pain point: spending 4-8 hours/day on page management, missing leads, inconsistent response times

US Market Opportunity

- ~33.2 million small businesses in the US (SBA, 2024)
- 93% of SMBs use social media for marketing
- Average cost of a social media manager: $4,000-6,000/month
- BizPilot at $200-500/month = massive cost savings
- Competitors (ManyChat, Tidio, Intercom) are rule-based or expensive at scale

Goal

- **100 customers within 6 months** (starting March 2026)
- Initial pilots with Vietnamese SMEs (existing network) to validate product
- Shift primary acquisition to US market by Month 3
- Solo dev (Thang) + Claude Code as co-builder

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2. Product Definition

What Each Business (Customer) Gets

1. **1 Dedicated AI Agent** - unique brain, knowledge, and brand voice
2. **Multi-channel connections** (market-dependent):
    - Facebook Messenger (inbox auto-reply) - all markets
    - Facebook/Instagram Comments (monitor + reply) - all markets
    - Instagram DM (inbox auto-reply) - US priority
    - Website Chat widget (built-in OpenClaw web provider) - all markets
    - WhatsApp Business (built-in) - US, LATAM, Europe
    - Zalo OA (existing extension) - Vietnam only
    - Google Business Messages (future) - US priority
3. **Lead capture** - automatically collects customer info, saves to DB, notifies admin
4. **Content assistant** - drafts posts, admin approves via Telegram
5. **Daily reports** - sent via Telegram every evening
6. **Escalation** - detects complex issues, alerts admin
7. **Admin control** - manage everything via Telegram

Automation Levels (Honest Assessment)

Activity	Automated	Needs Human
Reply inbox FAQ (price, stock, shipping)	90%	10%
Reply comments (simple)	85%	15%
Hide spam/toxic comments	95%	5%
Draft content/captions	70%	30% (approval)
Post on schedule	90%	10%
Product photography/video	0%	100%
Handle complaints/refunds	20%	80%
Run ads/boost posts	30%	70%
Analytics reporting	90%	10%
Crisis management (bad PR)	0%	100%
KOL/wholesale partnerships	10%	90%
Update catalog (new products)	20%	80%

Overall: Agent automates ~60-65% of workload, reduces admin time from 8h to 2-3h/day

Positioning

- **DO NOT pitch**: "AI replaces your employee"
- **DO pitch**: "AI assistant that makes your admin 3-4x more productive"
- 1 person + BizPilot = productivity of 3-4 people

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3. Architecture

Core Platform

- **OpenClaw** (open-source) as agent runtime
- BizPilot = OpenClaw + Facebook extension + custom tools + multi-tenant config

System Architecture

```
GitHub (code)
  | push
  v
GitHub Actions (CI/CD)
  | auto deploy
  v
Railway (hosting)
  |
  |-- OpenClaw Gateway (Docker container)
  |   |
  |   |-- Agent "shop-a" --> FB Page A + Web Chat A + Zalo A
  |   |-- Agent "shop-b" --> FB Page B + Web Chat B
  |   |-- Agent "shop-c" --> FB Page C + Web Chat C + WhatsApp C
  |   |-- ... (10-15 agents per instance)
  |   |
  |   |-- Admin channels:
  |       |-- Telegram @admin_a (shop A owner)
  |       |-- Telegram @admin_b (shop B owner)
  |       |-- Telegram @admin_c (shop C owner)
  |
  |-- Supabase (external, managed)
  |   |-- PostgreSQL
  |   |   |-- tenants (business profiles)
  |   |   |-- products (per tenant, supports 10k+ rows)
  |   |   |-- leads (captured from all channels)
  |   |   |-- customers (cross-channel profiles)
  |   |   |-- analytics_events
  |   |   |-- conversations_log
  |   |-- Storage
  |   |   |-- product-images/{tenant_id}/
  |   |-- Row Level Security (automatic tenant isolation)
  |
  |-- Cloudflare
      |-- DNS (custom domains)
      |-- CDN (cache static assets)
```

```
      |-- Pages (web chat widget hosting)
      |-- R2 (backup/overflow storage)
```

Per-Business Architecture

```
┌──────────────────────────────────────────────────────┐
│                 1 Business = 1 Agent                 │
│                                                      │
│  ┌──────────────────────────────────────────────┐ │
│  │           Agent "Business Name"              │ │
│  │                                              │ │
│  │  Brain:     LLM (Claude/GPT)                 │ │
│  │  Knowledge: products, pricing, FAQ, policies  │ │
│  │  Voice:     unique brand tone & personality  │ │
│  │  Tools:     catalog, lead capture, analytics...  │ │
│  │  Memory:    conversation history, customer data  │ │
│  │  Cron:      content schedule, reports, monitors  │ │
│  └──────────┬──────────────────────────────┬──────┘ │
│             |                              |       |
│    ┌────────▼─────────┐            ┌──────────▼──┐ │
│    |  Customer-facing  |            |  Admin-facing  | |
│    |                  |            |               | |
│    |  Facebook Inbox   |            |  Telegram      | |
│    |  Facebook Comment |            |  - Approve     | |
│    |  Website Chat     |            |  - Reports     | |
│    |  Zalo OA          |            |  - Direct      | |
│    |  WhatsApp         |            |  - Alerts      | |
│    |  Instagram DM     |            |  - Override    | |
│    |                  |            |               | |
│    |  --> Reply        |            |  Or Web UI     | |
│    |  --> Capture lead |            |  (OpenClaw)    | |
│    |  --> Escalate     |            |               | |
│    └──────────────────┘            └──────────────┘ |
└──────────────────────────────────────────────────────┘
```

Multi-Tenant Model

```
1 Railway instance = 10-15 businesses
100 customers = 7-10 instances

Each business gets:
  - 1 OpenClaw Agent (isolated agentDir, model, tools, memory)
  - 1 set of channel connections (FB token, Zalo token, web chat embed)
  - Data isolated via Supabase RLS (tenant_id column)
  - Routing: channel account --> agent (binding-based)
```

Scaling Plan

Stage	Customers	Infrastructure	Monthly Cost
Pilot	1-5	1 Railway instance	~$20

Growth	5-20	1 Railway instance (upgraded)	~$40-60
Scale	20-50	2-3 instances	~$100-150
Target	50-100	5-7 instances	~$200-350

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4. Tech Stack (Decided)

Core Stack

Layer	Technology	Claude Code Access	Why This Choice
Agent platform	OpenClaw	CLI `openclaw`	Multi-agent, multi-channel, plugin system, open-source
Database	Supabase (PostgreSQL)	MCP server	Direct query/migrate/manage, RLS, Storage, generous free tier
Image storage	Supabase Storage	MCP/API	Built-in CDN, RLS-protected, integrated with DB
Hosting	Railway	CLI `railway`	Deploy from terminal, log streaming, auto-scaling
DNS/CDN	Cloudflare	CLI `wrangler`	Domain management, SSL, edge caching
Source control	GitHub	CLI `gh`	PRs, issues, CI/CD, collaboration
Monitoring	Better Stack	CLI + webhooks	Uptime monitoring, alerting
CI/CD	GitHub Actions	`gh` CLI	Auto deploy on push, test pipeline

MCP Servers (Claude Code Integration)

```
{
  "mcpServers": {
    "supabase": {
      "command": "npx",
      "args": ["-y", "@supabase/mcp-server-supabase@latest"],
      "env": { "SUPABASE_ACCESS_TOKEN": "..." }
    },
    "postgres": {
      "command": "npx",
      "args": ["@modelcontextprotocol/server-postgres", "postgresql://..."]
    }
  }
}
```

OpenClaw Platform Capabilities (Verified from Codebase)

Capability	Status	Details
Multi-agent on 1 gateway	Ready	Unlimited agents, each with isolated agentDir
Per-agent system prompt, model, tools	Ready	Full isolation across agents
Channel-to-agent routing	Ready	Binding-based: channel + accountId --> agentId
Multi-account per channel type	Ready	`accounts: { "shop-a": {...}, "shop-b": {...} }`
Per-agent cron jobs	Ready	Schedule, channel delivery, failure alerts
Custom tools via plugins	Ready	`registerTool()` with full agent/session context
Webhook HTTP endpoints	Ready	`registerHttpRoute()` + rate limiting + signature validation
Per-agent memory	Ready	SQLite + vector search, isolated per agent
Web Chat	Ready	Built-in web provider (`src/provider-web.ts`)
Zalo channel	Ready	`extensions/zalo/`
WhatsApp	Ready	Built-in (Baileys library)
Telegram	Ready	Built-in (grammY library)
Facebook Messenger	**Needs Building**	Follow pattern from `extensions/zalo/`

Rejected Alternatives

Option	Reason for Rejection
Cloudflare MoltWorker	Proof-of-concept only, not production-ready, channels need always-on process
AWS (EC2/ECS)	Over-engineered for solo dev, higher cost, more ops overhead
Self-managed PostgreSQL	Unnecessary ops burden; Supabase provides managed DB + extras

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5. Data Architecture

Supabase Schema (Planned)

```sql
-- Tenant (business) management
CREATE TABLE tenants (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  name text NOT NULL,
```

```sql
  slug text UNIQUE NOT NULL,
  owner_name text,
  owner_phone text,
  owner_telegram_id text,
  agent_id text NOT NULL,         -- OpenClaw agent ID
  plan text DEFAULT 'basic',      -- basic | standard | premium
  status text DEFAULT 'trial',    -- trial | active | suspended | cancelled
  config jsonb DEFAULT '{}',      -- brand voice, policies, custom settings
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz DEFAULT now()
);

-- Product catalog
CREATE TABLE products (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  name text NOT NULL,
  description text,
  price numeric NOT NULL,
  sale_price numeric,
  category text,
  tags text[] DEFAULT '{}',
  image_urls text[] DEFAULT '{}',
  stock integer DEFAULT 0,
  status text DEFAULT 'active',   -- active | out_of_stock | hidden
  metadata jsonb DEFAULT '{}',    -- variants, specs, custom fields
  created_at timestamptz DEFAULT now(),
  updated_at timestamptz DEFAULT now()
);

-- Lead capture
CREATE TABLE leads (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  name text,
  phone text,
  email text,
  source text NOT NULL,           -- facebook | web | zalo | whatsapp | instagram
  channel_user_id text,           -- platform-specific user ID
  interest text,                  -- what product/service they asked about
  conversation_summary text,      -- AI-generated summary of conversation
  status text DEFAULT 'new',      -- new | contacted | converted | lost
  metadata jsonb DEFAULT '{}',
  created_at timestamptz DEFAULT now()
);

-- Customer profiles (cross-channel unified)
CREATE TABLE customers (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  name text,
  phone text,
```

```sql
  email text,
  channel_ids jsonb DEFAULT '{}', -- { "facebook": "id", "zalo": "id", ... }
  total_orders integer DEFAULT 0,
  total_spent numeric DEFAULT 0,
  tags text[] DEFAULT '{}',
  notes text,
  last_contact_at timestamptz,
  created_at timestamptz DEFAULT now()
);

-- Analytics events
CREATE TABLE analytics_events (
  id uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  tenant_id uuid NOT NULL REFERENCES tenants(id),
  event_type text NOT NULL,       -- inbox_reply | comment_reply | lead_captured |
...
  channel text,                   -- facebook | web | zalo | whatsapp
  metadata jsonb DEFAULT '{}',
  created_at timestamptz DEFAULT now()
);

-- Row Level Security (all tables)
-- ALTER TABLE products ENABLE ROW LEVEL SECURITY;
-- CREATE POLICY "tenant_isolation" ON products
--   USING (tenant_id = current_setting('app.tenant_id')::uuid);
```

Data Responsibility Split

```
OpenClaw Memory (built-in)              Supabase (external DB)
------------------------------          ----------------------------
- Brand voice & tone guide              - Products (structured, queryable)
- FAQ & common responses                - Leads (captured contacts)
- Policies & guidelines                 - Customers (cross-channel profiles)
- Conversation history (sessions)       - Analytics events
- Agent bootstrap context               - Product images (Storage)
                                        - Tenant configuration
Suitable for:                           Suitable for:
  <= few hundred documents                10k+ rows per tenant
  Semantic/keyword search                 Exact queries, filtering, aggregation
  Unstructured knowledge                  Structured business data
```

Product Catalog Performance

Scale	DB Approach	Expected Latency
<= 200 products	PostgreSQL (basic queries)	< 100ms
200-2,000 products	PostgreSQL + B-tree indexes	< 100ms
2,000-10,000 products	PostgreSQL + pg_trgm (trigram search)	< 200ms
10,000+ products	PostgreSQL + pgvector (semantic search)	< 500ms

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6. Custom Tools (To Build)

Tool: product-search

```
Purpose: Search product catalog for customer inquiries
Input:   { query: string, category?: string, maxPrice?: number, minPrice?: number,
inStock?: boolean }
Output:  { products: [{ name, price, salePrice, stock, imageUrl, description }],
total: number }
Action:  Query Supabase products table with filters
```

Tool: save-lead

```
Purpose: Capture potential customer information during conversation
Input:   { name?: string, phone?: string, email?: string, source: string, interest:
string, conversationSummary: string }
Output:  { leadId: string, status: string }
Action:  INSERT into Supabase leads table + send notification to admin via Telegram
```

Tool: fb-comments

```
Purpose: Monitor and manage Facebook Page comments
Input:   { action: "list" | "reply" | "hide", postId?: string, commentId?: string,
text?: string }
Output:  { success: boolean, data: any }
Action:  Facebook Graph API comment operations
```

Tool: fb-publish

```
Purpose: Publish or schedule posts to Facebook Page
Input:   { message: string, imageUrl?: string, scheduledTime?: string }
Output:  { postId: string, url: string }
Action:  Facebook Graph API page publish
```

Tool: fb-insights

```
Purpose: Retrieve Facebook Page analytics
Input:   { period: "day" | "week" | "month", metrics?: string[] }
Output:  { reach: number, engagement: number, impressions: number, ... }
Action:  Facebook Insights API
```

Tool: daily-report

```
Purpose: Generate and deliver daily business summary
Input:   { tenantId: string, period: "day" | "week" }
Output:  { formatted report text with metrics }
Action:  Aggregate from analytics_events + leads + products in Supabase
```

Tool: escalate

```
Purpose: Alert business admin about issues requiring human attention
Input:   { reason: string, customerInfo: object, urgency: "low" | "medium" | "high",
conversationSummary: string }
Output:  { notified: boolean }
Action:  Send formatted alert to admin's Telegram with full context and action link
```

Tool: check-order (Phase 2+)

```
Purpose: Look up order status for customer inquiries
Input:   { orderId?: string, phone?: string }
Output:  { orderStatus: string, tracking: string, estimatedDelivery: string }
Action:  Query external order/shipping system
```

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7. Message Flow Details

Flow 1: Customer Inbox Reply (Reactive)

```
Customer sends "How much is Product X?" on Messenger
  |
  v
Facebook Messenger extension receives webhook
  |
  v
Gateway routes to correct agent (by FB Page account ID)
  |
  v
Agent checks Memory (returning customer? load history)
  |
  v
Agent calls Tool: product-search { query: "Product X" }
  --> Returns: price, description, stock, image URL
  |
  v
Agent composes reply in brand voice:
  "Hi! Product X is currently 350,000 VND.
   [product image]
   We have free shipping for orders over 500,000 VND.
   Would you like to order?"
  |
  v
Reply sent via Messenger
  |
  v
Memory updated: customer asked about Product X, interested in price
```

Flow 2: Website Chat + Lead Capture (Reactive)

```
Visitor on website clicks chat widget
  |
  v
WebSocket connects to OpenClaw Gateway --> Routes to agent
  |
  v
Customer: "Do you ship to Da Nang?"
  |
  v
Agent calls Tool: product-search (checks shipping policy in memory)
  --> "Yes, shipping to Da Nang: 30k VND, 2-3 days"
  |
  v
Agent detects purchase intent, initiates lead capture:
  "I'd love to help you order! May I have your name and phone number?"
  |
  v
Customer provides info
  |
  v
Agent calls Tool: save-lead
  { name: "Nguyen Van A", phone: "0901234567",
    source: "web", interest: "Product X, shipping to Da Nang" }
  |
  v
Admin receives Telegram notification:
  "New lead from website:
   Nguyen Van A - 0901234567
   Interest: Product X, Da Nang shipping"
```

Flow 3: Comment Management (Reactive via Cron)

```
Cron job runs every 2 minutes: check new comments
  |
  v
Agent calls Tool: fb-comments { action: "list" }
  --> Returns new comments on recent posts
  |
  v
Agent classifies each comment:
  |-- [Info question] --> Reply with answer
  |-- [Complaint]     --> Reply + escalate to admin
  |-- [Spam/toxic]    --> Hide + report
  |-- [Positive]      --> React + thank reply
  |
  v
Example reply: "Still in stock! DM us for quick ordering"
```

Flow 4: Daily Report (Proactive via Cron)

```
Cron trigger: 9:00 PM daily
  |
  v
Agent calls Tool: daily-report { period: "day" }
  --> Aggregates: inbox count, comments, leads, page insights
  |
  v
Agent sends to admin via Telegram:
  "Daily Report 03/12:
   - Inbox: 47 messages (43 auto-replied, 4 escalated)
   - Web chat: 12 messages (10 auto-replied, 2 new leads)
   - Comments: 23 replied, 2 spam hidden
   - New leads today: 6 (3 FB, 2 Web, 1 Zalo)
   - Page reach: 2,300 (+15% vs yesterday)

   Pending items:
   - Customer A: product defect complaint (needs your response)
   - Customer B: wholesale inquiry"
```

Flow 5: Content Scheduling (Proactive via Cron)

```
Cron trigger: 8:00 AM daily
  |
  v
Agent checks content calendar + product catalog
  |
  v
Agent drafts post: caption + hashtags + selects product image
  |
  v
Sends draft to admin via Telegram:
  "Today's post draft:
   [Preview with image]
   Reply 'ok' to publish at 10 AM
   Reply 'edit ...' to modify"
  |
  v
Admin replies "ok"
  |
  v
Agent calls Tool: fb-publish { message, imageUrl, scheduledTime: "10:00" }
  --> Post published/scheduled on Facebook Page
```

Flow 6: Escalation (Auto-triggered)

```
Customer: "Your product gave me an allergic reaction! I want a refund!"
  |
  v
Agent detects: negative sentiment + complaint + refund request
  --> Exceeds agent's authority (per guidelines)
```

```
  |
  v
Agent replies to customer (stalling):
  "I'm very sorry about your experience.
   I've escalated this to our support team.
   Someone will contact you within 30 minutes."
  |
  v
Agent calls Tool: escalate {
  reason: "Product complaint + refund request",
  urgency: "high",
  customerInfo: { name, history: "VIP, 5 previous orders" }
}
  |
  v
Admin receives Telegram alert:
  "URGENT - Refund Request
   Customer: Nguyen Van A (Messenger)
   Issue: allergic reaction, requesting refund
   History: VIP customer, 5 orders, 2.5M VND total
   Direct link: [messenger conversation link]"
```

Flow 7: Cross-Channel Context

```
Day 1: Customer asks on website about Product X (price 350k)
  --> Agent remembers in memory

Day 3: Same customer messages on Facebook: "I asked about some cream before..."
  --> Agent searches memory by phone/email match
  --> "You asked about Product X (350k VND)!
       We actually have a 10% discount this week..."
```

Note: Cross-channel memory only works if the customer can be identified across channels (phone, email, or login). Anonymous web sessions remain separate.

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8. Admin Experience (Via Telegram)

What the business owner sees on Telegram

```
[Agent] Daily Report 03/12:
        - Inbox: 47 messages (43 auto, 4 escalated)
        - Web chat: 12 messages (2 new leads)
        - Leads today: 6 (3 FB, 2 Web, 1 Zalo)
        - Reach: 2,300 (+15% vs yesterday)

[Agent] New Lead - Website:
        Tran Thi B - 0912345678
        Interest: Skincare combo set
        Budget: ~500k VND
```

```
[Agent] Content draft for tomorrow:
        [Post preview with image]
        Reply 'ok' to publish at 8 AM
        Reply 'edit ...' to modify

[Admin] "reply to lead Tran Thi B that we have a 450k combo with free shipping"
[Agent] --> Sends message to customer on the original channel

[Admin] "what sold best this week?"
[Agent] "Skincare combo: 23 sets, revenue 10.3M VND
         Top channel: Facebook (15), Web (5), Zalo (3)"
```

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9. Pricing Model

Service Tiers (USD - US Market)

Tier	Scope	Monthly Fee
Starter	1 channel (FB or Web Chat) + inbox auto-reply + FAQ + escalation	$149/month
Growth	2-3 channels + comments + reports + lead capture	$299/month
Pro	All channels + content scheduling + analytics + priority support	$499/month

No setup fee for US market (SaaS model, self-service onboarding goal).

Service Tiers (VND - Vietnam Market)

Tier	Scope	Setup Fee (One-time)	Monthly Fee
Basic	Inbox auto-reply + FAQ + escalation	15-25M VND	3-5M VND
Standard	Basic + comments + reports + web chat + lead capture	25-40M VND	5-8M VND
Premium	Standard + content scheduling + multi-channel + analytics	40-70M VND	8-15M VND

Our Operating Costs (at 100 customers)

Service	Monthly Cost
Railway (hosting, 5-7 instances)	$200-350
Supabase (managed PostgreSQL + Storage)	$50-100
Cloudflare (DNS/CDN/Pages)	$0-5
LLM API (Claude Haiku for simple, Sonnet for complex)	$200-500
Total infrastructure	**~$500-1,000/month**
Revenue (100 customers x $300 avg)	**~$30,000/month**

Gross margin	**~96%**

Competitive Comparison (US Market)

Solution	Monthly Cost	AI Quality	Multi-Channel	24/7
Hire social media manager	$4,000-6,000	Human	Multi	No (business hours)
ManyChat Pro	$15-65	Rule-based flows	FB/IG only	Yes
Tidio+	$29-394	Basic AI + rules	Web + FB	Yes
Intercom	$39-139/seat	AI copilot	Web + email	Yes
Drift	$2,500+	AI chatbot	Web	Yes
BizPilot	**$149-499**	**LLM (contextual)**	**All channels**	**Yes**

Key differentiators vs US competitors:

- **vs ManyChat**: True AI understanding vs rigid flow builders; multi-channel beyond just FB/IG
- **vs Tidio/Intercom**: Full page management (comments, content, reports), not just chat widget
- **vs Drift**: 10x cheaper, designed for SMBs not enterprise
- **vs hiring**: 10-30x cheaper than a social media manager, works 24/7, never calls in sick

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10. Build Roadmap (6 Months)

Phase 1: MVP (Month 1 - 4 weeks)

Goal: 2-3 pilot customers (VN network, free or discounted)

```
Week 1-2: Facebook Messenger Extension
  - Webhook receiver (Graph API verification + callback handling)
  - Send messages (text, images, quick replies)
  - Multi-account resolution (multiple FB Pages)
  - Onboarding flow (Page token setup)
  - Reference: extensions/zalo/ pattern

Week 2-3: Core Tools + Instagram DM
  - product-search tool (Supabase query)
  - save-lead tool (capture + Telegram/email notification)
  - escalate tool (alert admin)
  - Instagram DM support (shares Graph API with Facebook)

Week 3-4: Integration & Testing
  - Multi-tenant OpenClaw config setup
  - Web chat embed script (JS snippet for customer websites)
  - Supabase schema deployment + seed data
```

```
- End-to-end testing with real FB Page + Instagram
- Deploy to Railway
```

Phase 2: Stabilize + US Prep (Month 2 - 3 weeks)

Goal: 5-10 customers (mix of VN pilots + first US customers)

```
- Facebook/Instagram comment monitoring (Graph API polling via cron)
- Daily report tool (Insights API --> format --> delivery)
- Landing page + sign-up flow (bizpilot.ai or similar)
- Stripe billing integration (US market requires self-service payment)
- Automated onboarding script (faster new customer setup)
- Bug fixes from pilot feedback
```

Phase 3: US Market Launch (Month 3 - 3 weeks)

Goal: 15-25 customers (US-focused acquisition)

```
- Self-service onboarding (connect FB Page, import products, go live)
- Content scheduling (cron + Graph API publish)
- WhatsApp Business API integration (important for US SMBs)
- Cross-channel customer matching (phone/email lookup)
- Improved lead capture conversational flow
- Marketing: Product Hunt launch, SMB communities, cold outreach
```

Phase 4: Growth (Month 4 - 4 weeks)

Goal: 30-50 customers

```
- Web dashboard for customers (view leads, analytics, manage catalog)
- CSV/Google Sheet product catalog upload
- Multi-language support (LLM handles natively, but UI/onboarding i18n)
- Admin mobile notifications (beyond Telegram - email, push)
- Zalo OA for VN customers (existing extension, configure per tenant)
```

Phase 5: Scale (Month 5 - 3 weeks)

Goal: 60-80 customers

```
- Google Business Messages channel (US priority)
- Multi-instance deployment automation
- Performance optimization (LLM cost, response latency)
- Monitoring & alerting (Better Stack)
- SOC 2 / security compliance prep (US enterprise customers)
```

Phase 6: Polish & Expand (Month 6 - 3 weeks)

Goal: 100 customers

```
- Customer referral program
- Case studies from successful US customers
- Advanced analytics dashboard
```

```
  - API for third-party integrations (Shopify, WooCommerce, etc.)
  - Explore: LATAM market (Spanish), Europe (multi-language)
```

---

11. Customer Onboarding Flow

Per-Customer Setup Process

```
Day 1: Contract signed
  |
  v
Day 2-3: Technical Setup (Claude Code assisted)
  |- Create new OpenClaw agent (config + agentDir)
  |- Connect Facebook Page (obtain Page Access Token)
  |- Embed web chat widget on customer's website (1 line of JS)
  |- Connect Zalo OA / WhatsApp (if applicable)
  |- Connect admin's Telegram account
  |- Import product catalog (CSV upload to Supabase)
  |- Customize system prompt (brand voice, FAQ, policies)
  |
  v
Day 4-5: Testing & Tuning
  |- Test conversations on every connected channel
  |- Tune reply quality (add FAQ entries, adjust tone)
  |- Train admin on Telegram commands
  |
  v
Day 6: Go Live
```

After product layer stabilizes: new customer onboarding takes only 2-3 days.

---

12. Claude Code Collaboration Workflow

How We Work Together

Thang directs, Claude Code executes via CLI and MCP:

```
# Adding a new customer
Claude Code:
  1. MCP --> Supabase: INSERT INTO tenants (name, agent_id, ...) VALUES (...)
  2. Edit openclaw config.json5 (add agent + channel routing)
  3. $ railway up (redeploy gateway)
  4. Verify: test webhook, send test message
  --> Done in ~10 minutes

# Debugging a customer issue
Claude Code:
  1. $ railway logs --filter "shop-abc"
  2. MCP --> Supabase: SELECT * FROM products WHERE tenant_id = 'shop-abc'
  3. Identify issue (wrong data, prompt issue, etc.)
  4. Fix code or config
```

```
  5. $ railway up
  --> Done

# Generating weekly business report
Claude Code:
  1. MCP --> Supabase:
     SELECT tenant_id, COUNT(*) as leads,
            COUNT(DISTINCT source) as channels
     FROM leads
     WHERE created_at > now() - interval '7 days'
     GROUP BY tenant_id
  2. Format and present results
  --> Instant

# Deploying new features
Claude Code:
  1. Write code (tools, extensions, configs)
  2. $ git add && git commit && git push
  3. GitHub Actions auto-deploys to Railway
  4. Verify: $ railway logs
  --> Continuous delivery
```

Required MCP Server Configuration

```
{
  "mcpServers": {
    "supabase": {
      "command": "npx",
      "args": ["-y", "@supabase/mcp-server-supabase@latest"],
      "env": {
        "SUPABASE_ACCESS_TOKEN": "<your-access-token>"
      }
    },
    "postgres": {
      "command": "npx",
      "args": [
        "@modelcontextprotocol/server-postgres",
        "postgresql://<user>:<pass>@<host>:5432/<db>"
      ]
    }
  }
}
```

Note: Thang already has `@modelcontextprotocol/server-postgres` installed globally.

---

13. Risk Assessment & Mitigation

Risk	Impact	Probability	Mitigation Strategy

Facebook API changes or restrictions	High	Medium	Abstraction layer between agent tools and Graph API; monitor changelog; version-pin API calls
LLM hallucination (wrong price/product info)	High	Medium	Product catalog tool is single source of truth; system prompt guard rails; never generate prices from memory
Customer expects 100% accuracy	Medium	High	Clear SLA: ~90% automated, ~10% escalated; set expectations during onboarding
LLM API costs spike with volume	Medium	Medium	Use Claude Haiku for simple queries, Sonnet for complex; implement response caching; set per-tenant rate limits
Competition (other AI chatbot services)	Medium	Medium	Move fast; focus on vertical niches; accumulate domain knowledge; build switching costs via data
Railway/Supabase downtime	Medium	Low	Docker-based (portable to any cloud); Supabase has backups; Railway has multi-region
Tenant data leakage	High	Low	Supabase RLS on all tables; per-agent isolation in OpenClaw; audit logging; regular security review
Solo dev burnout / bottleneck	High	Medium	Claude Code handles routine ops; automate onboarding; prioritize self-service features (Phase 4)

---

14. Key Decisions Log

Date	Decision	Reasoning
2026-03-12	Project name: **BizPilot**	Broad scope (not just pages), professional, works internationally
2026-03-12	Platform: **OpenClaw**	Open-source, proven multi-agent + multi-channel + plugin architecture
2026-03-12	Database: **Supabase PostgreSQL**	MCP server support, RLS for tenant isolation, Storage for images, generous free tier
2026-03-12	Hosting: **Railway**	CLI-based deployment, Docker support, auto-scaling, log streaming
2026-03-12	Multi-tenant approach: **shared gateway**	10-15 agents per instance, scale by adding instances
2026-03-12	Product catalog in **external DB**	OpenClaw memory system not suitable for 10k+ structured product records

2026-03-12	**Rejected** Cloudflare MoltWorker	Proof-of-concept only, not production-ready, channels need always-on processes
2026-03-12	Pricing US: **$149-499/month** (SaaS)	10-30x cheaper than hiring, premium over rule-based bots, no setup fee
2026-03-12	Pricing VN: **3-15M VND/month** + setup fee	Localized pricing for Vietnam market
2026-03-12	Target: **100 customers in 6 months**	Aggressive but achievable with product-led approach and low marginal cost per customer
2026-03-12	Positioning: **force multiplier**	Not replacing people, making people 3-4x more productive; honest about limitations
2026-03-12	**Primary market: US SMEs**	Largest market, highest willingness to pay, SaaS-friendly; VN as pilot/secondary market
2026-03-12	**Multi-market architecture**	Language-agnostic agent (LLM handles any language), market-specific channel mix

---

15. Team

Member	Role	Capabilities
Thang	Solo dev, founder	Product vision, engineering, sales, customer relationships
Claude Code	AI co-builder	Code, deploy, debug, query DB, manage infra via CLI/MCP, available 24/7

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16. References

Resource	URL / Path
OpenClaw repository	https://github.com/openclaw/openclaw
OpenClaw documentation	https://docs.openclaw.ai
Zalo extension (reference pattern)	`openclaw/extensions/zalo/`
Facebook Graph API docs	https://developers.facebook.com/docs/graph-api
Facebook Messenger Platform	https://developers.facebook.com/docs/messenger-platform
Supabase documentation	https://supabase.com/docs
Railway documentation	https://docs.railway.app
Cloudflare Workers/Pages	https://developers.cloudflare.com
Better Stack	https://betterstack.com/docs

Appendix A: OpenClaw Codebase Key Paths

Area	Path	Relevance
Agent config types	`src/config/types.agents.ts`	Agent schema definition
Routing engine	`src/routing/resolve-route.ts`	Message-to-agent routing logic
Channel plugin types	`src/channels/`	Channel adapter interfaces
Plugin registration	`src/plugins/types.ts`	`registerTool()`, `registerHttpRoute()`, etc.
Cron system	`src/cron/types.ts`, `src/cron/service.ts`	Scheduled task execution
Memory system	`src/memory/manager.ts`, `src/memory/types.ts`	Per-agent memory with vector search
Web provider	`src/provider-web.ts`	Built-in web chat
Tool truncation limits	`src/agents/pi-embedded-runner/tool-result-truncation.ts`	Max 400K chars per tool result
Bootstrap context limits	`src/agents/pi-embedded-helpers/bootstrap.ts`	20K chars/file, 150K total
Zalo extension	`extensions/zalo/`	**Primary reference for Facebook extension**
Plugin SDK	`src/plugin-sdk/index.ts`	446 exports available to extensions

Appendix B: Glossary

Term	Definition
Agent	An OpenClaw AI agent instance configured for one business
Tenant	A business customer using BizPilot
Channel	A messaging platform (Facebook, Zalo, Web, WhatsApp, etc.)
Binding	A routing rule mapping a channel account to an agent
Escalation	Transferring a conversation from AI agent to human admin
Lead	A potential customer whose contact info has been captured
Tool	A function the AI agent can call (search products, save leads, etc.)
Cron	A scheduled task that runs automatically (reports, comment monitoring)
Gateway	The OpenClaw server process that manages all agents and channels

RLS	Row Level Security - Supabase feature for automatic tenant data isolation

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This document is updated whenever important decisions, changes, or milestones occur. Last update: 2026-03-12 - Initial version